CS2041 Data Structures - Course Guidelines (Sample Document)

1. Course Overview

CS2041 is a 4-credit core course with 3 lecture hours and 2 lab hours per week.
Prerequisite: CS1010 Programming Fundamentals with a minimum grade of C.

2. Assessment Scheme

Continuous Internal Evaluation (CIE) carries 50 marks and the Semester End Exam (SEE) carries 50 marks.
CIE consists of two internal tests of 15 marks each, a mini project of 15 marks and lab work of 5 marks.
A student must score at least 40 percent in SEE and 40 percent overall to pass the course.
Students with less than 75 percent attendance are not permitted to write the SEE.

3. Mini Project

The mini project is done in teams of at most 3 students.
Topics must be approved by the course instructor by the end of week 4.
The final report and demo are due in week 13, followed by a 10 minute viva.

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4. Late Submission and Academic Integrity

Lab records submitted late lose 10 percent of the marks per day, up to a maximum of 3 days. Submissions more than 3 days late receive zero marks unless a medical certificate is produced. Copying code from another student or from the internet without citation is treated as malpractice. Malpractice in an internal test results in zero marks for that test.

5. Syllabus Units

Unit 1: Arrays, linked lists and complexity analysis.
Unit 2: Stacks, queues and their applications.
Unit 3: Trees, binary search trees and heaps.
Unit 4: Graphs, BFS, DFS and shortest paths.
Unit 5: Hashing, sorting and searching techniques.

6. Textbook

Primary text: Data Structures and Algorithm Analysis in C by Mark Allen Weiss, 2nd edition.